



SQL Analysis: IMDB Movies

Uncovering correlations between movie runtime and audience ratings through SQL-driven analysis (2010–2019).



What We'll Cover

1. Objective and toolstack
2. Strategy and method
3. What the data shows
4. Findings & Takeaways



Objective

- To explore the relationship between **movie runtime** and **audience ratings** using SQL, Pandas, and Seaborn in order to identify trends that could support evidence-based decision-making in the film industry.
- **Data Source:** IMDB Movies Database (2010–2019)
- **Tools Used:** SQL, SQLite, Pandas, Seaborn, Matplotlib

Initial Exploration Findings

SQL Exploration

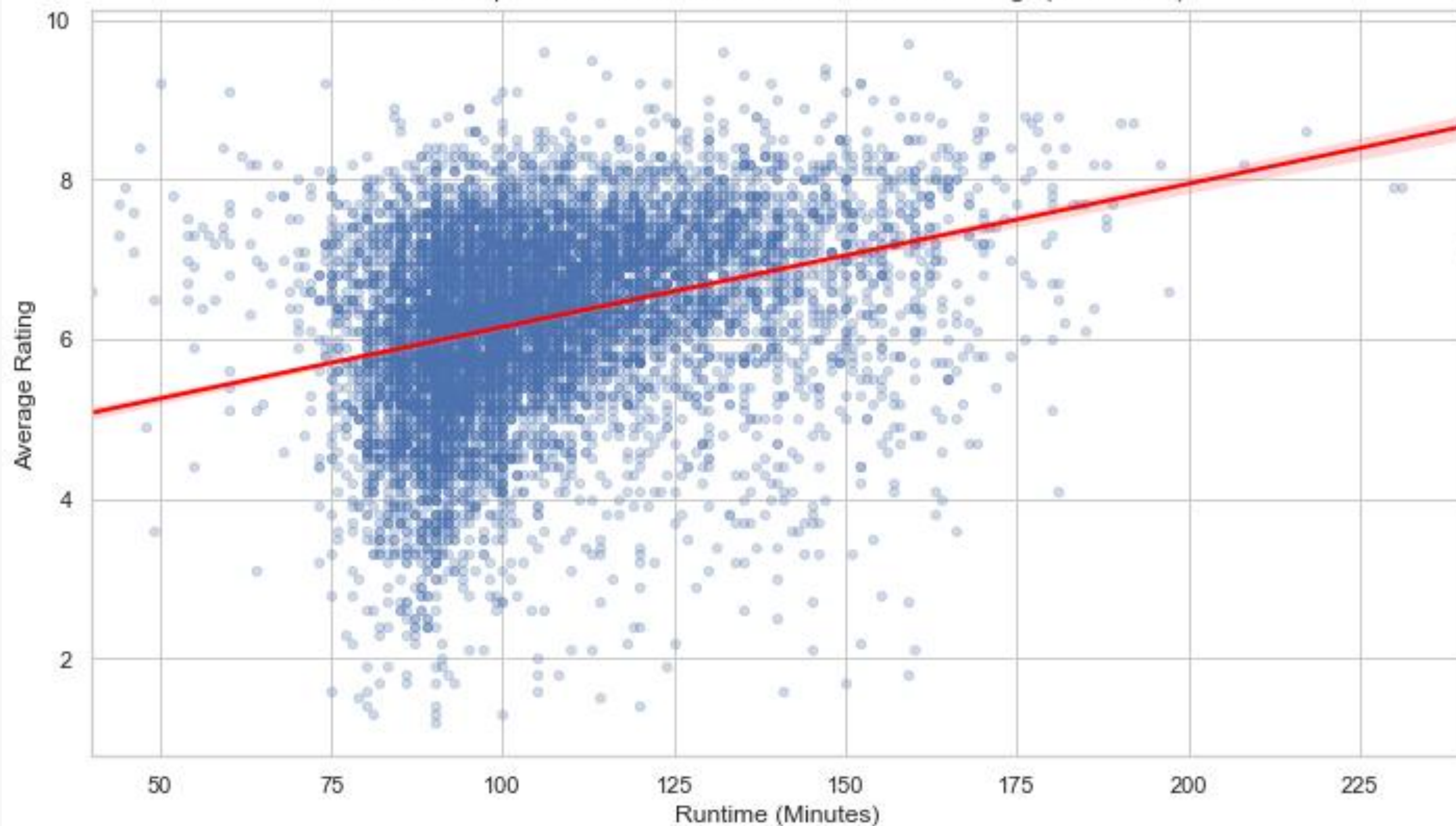
- Joined the **movie_basics** and **movie_ratings** tables using **movie_id**.
- Restricted the analysis to movies released between **2010 and 2019**.
- Excluded records with missing runtime and audience rating values.
- Filtered the dataset to include movies with runtimes between **40 and 240 minutes** to remove unrealistic values.
- Movies with **fewer than 1,000 audience** votes were excluded to improve the reliability of the ratings and reduce the influence of movies with limited audience feedback.
- Used a **scatter plot with a regression (trend) line** to examine the relationship between movie runtime and audience ratings.



✕ Business Question

Does *movie runtime* influence *audience ratings*, and is there an optimal runtime associated with higher-rated movies?

Relationship Between Movie Runtime and Audience Ratings (2010–2019)





Key Findings

- Most movies released between **2010 and 2019** have runtimes ranging from **80 to 140 minutes**.
- Audience ratings are concentrated between **5.5 and 7.5**, regardless of movie length.
- The regression line is **nearly flat with a slight downward slope**, indicating **little to no meaningful relationship** between runtime and audience ratings.
- Both highly rated and poorly rated movies occur across nearly all runtime lengths.
- The wide dispersion of data points suggests that **movie runtime is not a strong predictor of audience ratings**.
- Other factors—such as **genre, storyline, cast, director, production quality, and audience preferences**—are likely to have a much greater influence on how audiences rate a movie.



Business Insight

- This exploratory analysis found little evidence that movie runtime has a meaningful influence on audience ratings.
- Although moderately **longer movies are common**, the wide variation in ratings suggests that other factors—such as **genre**, **storytelling**, **cast**, and **production quality**—are more influential.
- Future analyses should incorporate these variables to better understand audience preferences.



Thank You

Questions welcome

Gabriel Manthi · July 15, 2026